

Mohammadreza Nemati

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EXPERIENCE

AI Engineer

05/2025 - Present

Volkswagen Group of America

Belmont, CA

- Developed long-horizon planning workflows for multi-agentic systems, including dynamic UI-tree planning, agentic RAG, and task decomposition for multimodal HMI automation.
- Architected production-grade multimodal inference infrastructure for agentic automotive HMI workflows, designing evaluation frameworks for latency, throughput, task accuracy, and serving reliability with vLLM.
- Fine-tuned multimodal foundation models using SFT, DPO, and GRPO for agentic reasoning, grounding, and task execution; applied knowledge distillation with quantization-aware training to compress VLMs for constrained hardware deployment.
- Pruned Qwen2.5-VL for autonomous-driving trajectory prediction, reducing parameters from 3.75B to 1.4B for deployment under edge compute and memory constraints.

Artificial Intelligence Research Assistant

08/2022 - 06/2025

Case Western Reserve University

Cleveland, Ohio

- Improved LLM KV-cache quantization by analyzing key/value norm imbalance, achieving $2\times$ memory reduction with $>94\%$ accuracy retention across models up to 70B parameters, directly optimizing inference memory efficiency for large-scale serving [\[paper link\]](#)
- Developed LIT-LVM using latent variable models with structured regularization and GNN-based higher-order interactions for biomedical data, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [\[paper link\]](#)
- Developed DEASurv, a deep survival model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve multi-modal biological survival prediction on large-scale medical datasets

Software Engineer - AI/ML Intern

02/2024 - 07/2024

Sherwin-Williams

Cleveland, Ohio

- Designed and deployed real-time inference pipeline for vision transformers (SAM) with client-backend architecture, optimizing model latency and achieving 7% IoU improvement through inference optimization techniques.
- Built multimodal system integrating vision-language models (LLaVA) with vision transformers, implementing cross-modal alignment for natural language-driven visual grounding and object segmentation.

Data Science Research Intern

05/2023 - 09/2023

Beyond Limits AI

Glendale, California

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 using PyTorch distributed training with FSDP/DDP for multi-GPU optimization and quantization for efficient deployment.

Machine Learning Research Assistant

08/2018 - 07/2022

University of Toledo

Toledo, Ohio

- Developed large-scale ML pipelines over 500,000+ records in PyTorch for preprocessing, distributed data handling, model development, and evaluation

SKILLS

- **Programming Languages:** Python, C++
- **Machine Learning:** Multimodal Foundation Models, Transformer Architectures, LLM/VLM Fine-tuning, Knowledge Distillation, Vision Transformers, Reinforcement Learning, Graph Neural Networks
- **ML Infrastructure:** Distributed Training (FSDP/DDP), Data Pipelines, Distributed Inference, Model Serving & Deployment, Profiling & Optimization
- **Frameworks & Libraries:** PyTorch, vLLM, LangChain/LangGraph, PyTorch Geometric, NumPy, Pandas
- **Inference Optimization:** CUDA, Quantization, Distillation, Continuous Batching, Pruning
- **Agentic Systems:** Long-Horizon Planning, Agentic RAG, Dynamic UI-Tree Planning, Task Decomposition
- **DevOps & Tools:** Docker, Flask, Redis, SQL, Git

EDUCATION

Case Western Reserve University

Cleveland, Ohio

Doctorate (PhD), Computer Science

08/2022 - 05/2025

University of Toledo

Toledo, Ohio

Masters (MS), Computer Science

08/2018 - 08/2020

Shiraz University

Shiraz, Fars

Bachelors (BS), Electrical Engineering

08/2012 - 05/2017